MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG2DSCSTA203- Statistical Research Techniques Using Softwares

Duration: 1Hour 30 Minutes **Maximum: 50 Marks**

PART A (Short answer type questions)

Answer any SEVEN questions. Each question Carries TWO marks

1. Define research design. [U] [1]
2. Distinguish between primary data and secondary data. [K] [1]
3. What is meant by literature review in a research process? [U] [1]
4. Define statistic with example. [U] [1]
5. Distinguish between simple and composite hypothesis. [K] [1]
6. What are the two types of errors in a statistical test procedure? [K] [1]
7. What are the assumptions used in t-test? [U] [3]
8. Distinguish between parametric and non parametric tests. [U] [2]
9. What is meant by contingency table? [K] [3]
10. Give any two applications of statistical tests in real life? [K] [4]

PART B (Short answer type questions)

Answer any FOUR questions. Each question Carries SIX marks

11. Distinguish between qualitative and quantitative research with examples. [K] [1]
12. Explain any three sampling techniques. [U] [1]
13. Explain different steps in a statistical test procedure. [U] [1]
14. A sample of 6,400 Englishmen has a mean height of 170 cms and a standard deviation of 6.4 cms, while a simple sample of 1600 Australians has a mean height of 172 cms and a standard deviation of 6.3 cms. Do the data indicate that Australians are on the average taller than Englishmen at 5% level of significance? [A] [2]
15. Explain one way ANOVA. [U] [3]
16. (a) Explain Wilcoxon signed rank test. State clearly the assumptions.

    (b) A high school track and field coach is interested to know whether the new training program improved his athletes 400 meter sprint time and decided to

conduct an analysis. He obtained the permission of 7 athletes to record their sprint times both before and after his new training program which are shown below. Use appropriate non-parametric test to check whether these scores provide evidence of an increase in median sprint time at 5% level of significance?

Athlete	1	2	3	4	5	6	7
Sprint Time (Before)	63	61	62	58	59	56	61
Sprint time (After)	58	57	59	57	58	55	55

[An] [4]

PART C (Short answer type questions)

Answer any ONE questions. Each question Carries TWELVE marks

17. Explain different steps in writing a research proposal. [U] [1]
18. Following data represents the experience and annual income of 400 employees in a company. Test whether there is any association between length of service of employees and their annual income.

Experience (in years)	Annual Income (in Dollars)		
	Less than 40000	40000-75000	More than 75000
Less than 5	50	75	25
5-10	25	50	25
10 or more	25	75	50

[An] [4]